

RLC Rectifier for a Micro Hydro Generator-Powered LED Circuit

Abstract

This report presents the design and simulation of an AC to DC rectifier for powering three parallel green LEDs from a 5 V peak, 20 Hz micro hydro generator. An LC filter ($L = 400$ mH, $C = 22$ μ F) with 100 Ω current limiting resistors was selected from available components and simulated using 4th-order Runge-Kutta (RK4) in Python. The LC configuration achieved the lowest ripple (0.759 V, 32.8%) with LED current safely below 20 mA (max 7.58 mA).

Introduction

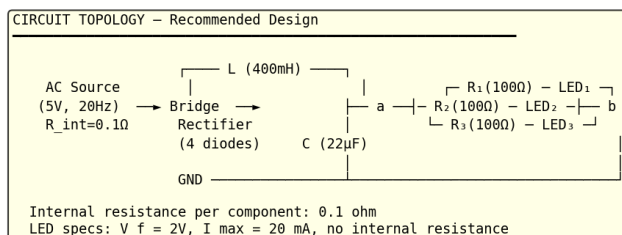
Converting AC to DC is essential for powering LEDs from an AC generator. This project uses a micro hydro generator (5 V peak, 20 Hz) to power three green LEDs ($V_f = 2$ V, $I_{max} = 20$ mA) in parallel. The low frequency (20 Hz) challenges conventional filtering.

The objectives are: (1) design an RLC rectifier from available components, (2) simulate using RK4, and (3) compare three filter configurations to justify the optimal design.

Methods

2.1 Circuit Model

The circuit consists of AC source ($V(t) = 5 \sin(2\pi \cdot 20 \cdot t)$ V, $R_{\text{int}} = 0.1 \Omega$), full-wave bridge rectifier (4 diodes, $V_d = 0.7$ V), LC filter ($L = 400$ mH series, $C = 22 \mu\text{F}$ parallel), and 3 parallel LED branches each with $R = 100 \Omega$. All components have 0.1Ω internal resistance. The 400 mH inductor and 22 μF capacitor were selected as the largest available values to maximize smoothing at 20 Hz. $R = 100 \Omega$ satisfies $R \geq (V_{\text{dc,max}} - V_{\text{LED}})/I_{\text{max}} = (3.6 - 2.0)/0.020 = 80 \Omega$ with safety margin.



2.2 Governing Equations

The LC filter is governed by two coupled ODEs:

- $L \cdot dI_L/dt = V_{rect}(t) - (R_{src} + R_{L,int}) \cdot I_L - V_C$ (KVL)
- $C \cdot dV_C/dt = I_L - I_{load}$ (KCL)

where $V_{rect}(t) = \max(|V_s(t)| - 2V_d, 0)$ and $I_{load} = 3 \cdot \max(V_C - V_{LED}, 0) / (R + R_{int})$

Constraints: $I_L \geq 0, V_C \geq 0$

2.4 Numerical Method

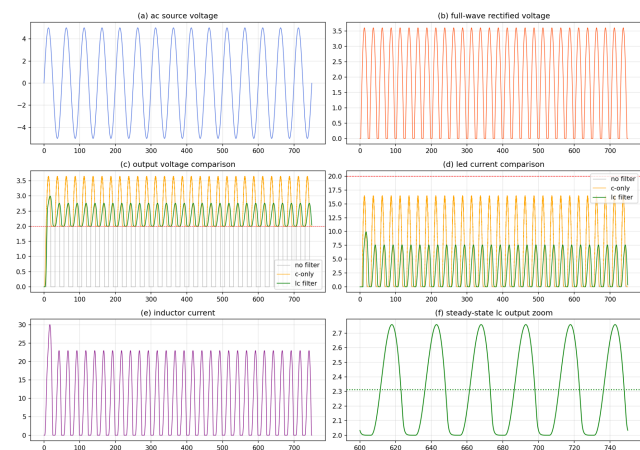
The system was solved using RK4, $y_{n+1} = y_n + (h/6)(k_1 + 2k_2 + 2k_3 + k_4)$, with $dt = 25 \mu s$ over 15 periods (750 ms). Three configurations were compared: no filter, C-only (22 μF), and LC (400 mH + 22 μF). The simulation was implemented in Python using NumPy and Matplotlib.

Results and Discussion

3.1 Filter Configuration Comparison

Configuration	V _{dc} [V]	Ripple [V]	Ripple [%]	I _{LED} avg [mA]	I _{LED} max [mA]
No Filter	1.598	3.595	224.9	5.50	15.94
C-only (22 μ F)	2.566	1.650	64.3	5.65	16.48
LC (400 mH + 22 μ F)	2.314	0.759	32.8	3.13	7.58

The LC filter reduces ripple by **78.9%** vs no filter and **54.0%** vs C-only, providing the smoothest output. All LED currents remain safely below 20 mA. The LC resonant frequency $f_0 = 1/(2\pi\sqrt{LC}) = 53.7$ Hz exceeds the 40 Hz ripple frequency, limiting attenuation, but the inductor's impedance and capacitor's charge storage still provide meaningful smoothing.

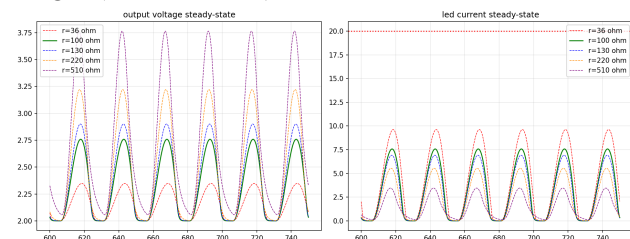


Simulation results: (a) AC source, (b) rectified voltage, (c) output voltage comparison of three filters, (d) LED current comparison, (e) inductor current, (f) steady-state LC output.

3.2 Current-Limiting Resistor Analysis

R [Ω]	I_LED avg [mA]	I_LED max [mA]	Safe
36	4.25	9.62	yes
100	3.13	7.58	yes
130	2.81	6.92	yes
220	2.19	5.55	yes
510	1.41	3.46	yes

R = 100 Ω was selected as the optimal trade-off between LED brightness (higher current) and safety margin (below 20 mA).



3.2 Error Analysis

Sources of error in this simulation include:

1. Real diodes have non-linear I-V characteristics, we used a constant $V_d = 0.7$ V

2. The diode switching and LED threshold introduce discontinuities that RK4 handles approximately
3. Real components have $\pm 5\text{--}20\%$ tolerance, affecting actual performance
4. V_{LED} and V_d vary with temperature

Conclusion

An LC filter ($L = 400$ mH, $C = 22$ μ F) with current-limiting resistors (3×100 Ω) successfully converts a 5 V peak, 20 Hz AC source to a DC supply for 3 parallel green LEDs. The steady state ripple is 0.759 V (32.8%) and peak LED current is 7.58 mA, well within the 20 mA safety limit. The computational approach using RK4 provides accurate transient and steady-state analysis, validated by comparison across three filter topologies

References

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2. Sedra, A.S. & Smith, K.C. (2020). *Microelectronic Circuits*, 8th ed. Oxford University Press
3. Chapra, S.C. & Canale, R.P. (2015). *Numerical Methods for Engineers*, 7th ed. McGraw-Hill.4g